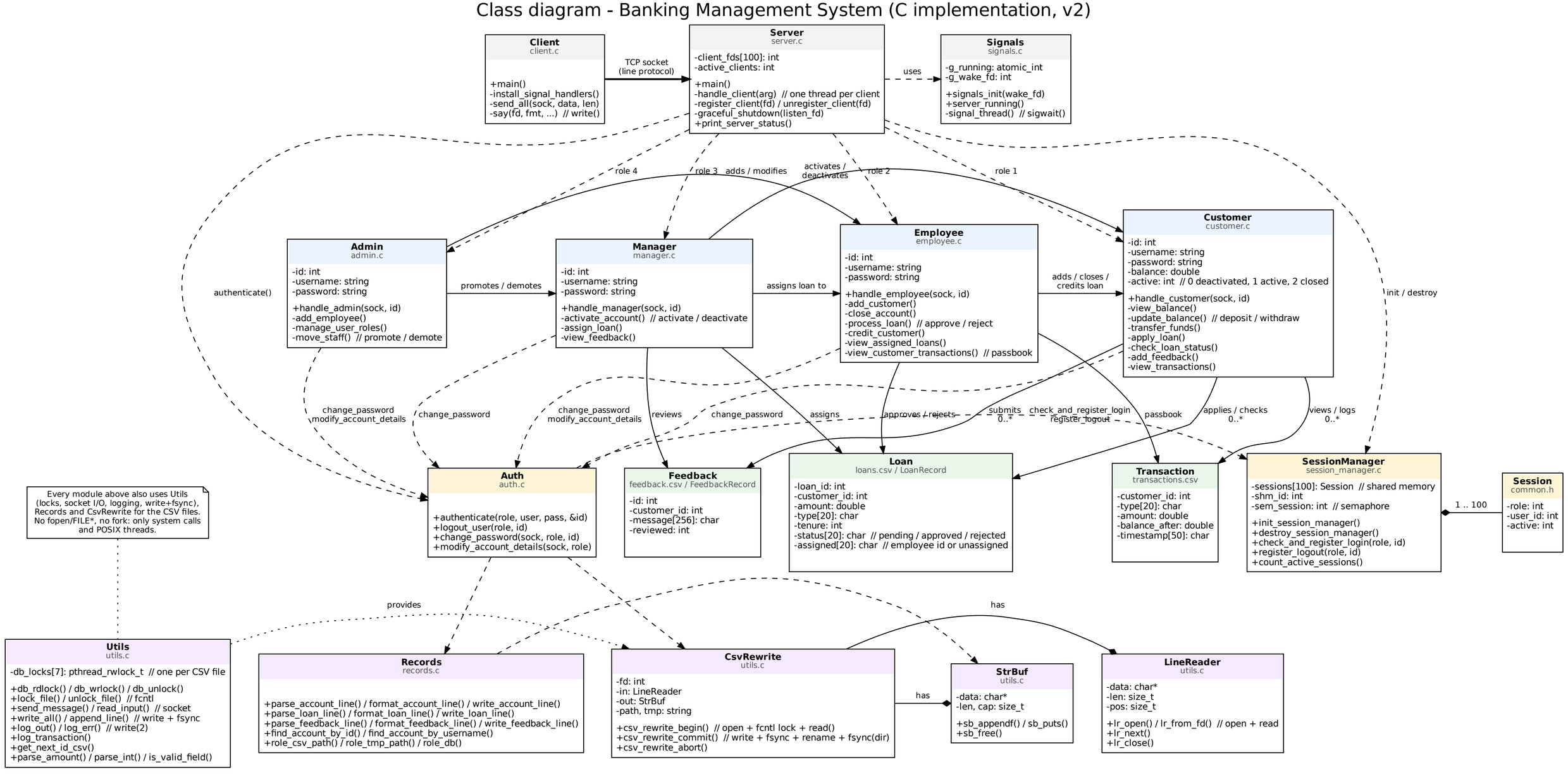
Class diagram - Banking Management System (C implementation, v2)
Client
client.c
+main()
-install_signal_handlers()
-send_all(sock, data, len)
-say(fd, fmt, ...) // write()
TCP socket
(line protocol)
Server
server.c
-client_fds[100]: int
-active_clients: int
+main()
-handle_client(arg) // one thread per client
-register_client(fd) / unregister_client(fd)
-graceful_shutdown(listen_fd)
+print_server_status()
uses
Signals
signals.c
-g_running: atomic_int
-g_wake_fd: int
+signals_init(wake_fd)
+server_running()
-signal_thread() // sigwait()
role 4
role 3
adds / modifies
activates /
deactivates
role 2
role 1
authenticate()
Admin
admin.c
-id: int
-username: string
-password: string
+handle_admin(sock, id)
-add_employee()
-manage_user_roles()
-move_staff() // promote / demote
promotes / demotes
Manager
manager.c
-id: int
-username: string
-password: string
+handle_manager(sock, id)
-activate_account() // activate / deactivate
-assign_loan()
-view_feedback()
assigns loan to
Employee
employee.c
-id: int
-username: string
-password: string
+handle_employee(sock, id)
-add_customer()
-close_account()
-process_loan() // approve / reject
-credit_customer()
-view_assigned_loans()
-view_customer_transactions() // passbook
adds / closes /
credits loan
Customer
customer.c
-id: int
-username: string
-password: string
-balance: double
-active: int // 0 deactivated, 1 active, 2 closed
+handle_customer(sock, id)
-view_balance()
-update_balance() // deposit / withdraw
-transfer_funds()
-apply_loan()
-check_loan_status()
-add_feedback()
-view_transactions()
init / destroy
change_password
modify_account_details
change_password
change_password
modify_account_details
reviews
change_password
assigns
approves / rejects
submits
0..*
check_and_register_login
register_logout
passbook
applies / checks
0..*
views / logs
0..*
Auth
auth.c
+authenticate(role, user, pass, &id)
+logout_user(role, id)
+change_password(sock, role, id)
+modify_account_details(sock, role)
Feedback
feedback.csv / FeedbackRecord
-id: int
-customer_id: int
-message[256]: char
-reviewed: int
Loan
loans.csv / LoanRecord
-loan_id: int
-customer_id: int
-amount: double
-type[20]: char
-tenure: int
-status[20]: char // pending / approved / rejected
-assigned[20]: char // employee id or unassigned
Transaction
transactions.csv
-customer_id: int
-type[20]: char
-amount: double
-balance_after: double
-timestamp[50]: char
SessionManager
session_manager.c
-sessions[100]: Session // shared memory
-shm_id: int
-sem_session: int // semaphore
+init_session_manager()
+destroy_session_manager()
+check_and_register_login(role, id)
+register_logout(role, id)
+count_active_sessions()
1 .. 100
Session
common.h
-role: int
-user_id: int
-active: int
Every module above also uses Utils
(locks, socket I/O, logging, write+fsync),
Records and CsvRewrite for the CSV files.
No fopen/FILE*, no fork: only system calls
and POSIX threads.
provides
has
has
Utils
utils.c
-db_locks[7]: pthread_rwlock_t // one per CSV file
+db_rdlock() / db_wrlock() / db_unlock()
+lock_file() / unlock_file() // fcntl
+send_message() / read_input() // socket
+write_all() / append_line() // write + fsync
+log_out() / log_err() // write(2)
+log_transaction()
+get_next_id_csv()
+parse_amount() / parse_int() / is_valid_field()
Records
records.c
+parse_account_line() / format_account_line() / write_account_line()
+parse_loan_line() / format_loan_line() / write_loan_line()
+parse_feedback_line() / format_feedback_line() / write_feedback_line()
+find_account_by_id() / find_account_by_username()
+role_csv_path() / role_tmp_path() / role_db()
CsvRewrite
utils.c
-fd: int
-in: LineReader
-out: StrBuf
-path, tmp: string
+csv_rewrite_begin() // open + fcntl lock + read()
+csv_rewrite_commit() // write + fsync + rename + fsync(dir)
+csv_rewrite_abort()
has
StrBuf
utils.c
-data: char*
-len, cap: size_t
+sb_appendf() / sb_puts()
+sb_free()
LineReader
utils.c
-data: char*
-len: size_t
-pos: size_t
+lr_open() / lr_from_fd() // open + read
+lr_next()
+lr_close()